

Summer Training Report.
Retail technical network
Telecom Egypt – We

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Content:

- ADSL, VDSL, Supervector Router
- DSL Technical Troubleshooting
- ONT, ONU FTTH Routers
- FTTH Technical Troubleshooting
- MIFI, Wingle, WEair Mobile Internet Devices & Troubleshooting

ADSL, VDSL, Supervector Router

Feature	ADSL Router	VDSL Router	SuperVector Router
Full Name	Asymmetric Digital Subscriber Line	Very-high-speed Digital Subscriber Line	Super Vectoring DSL
Download Speed	Up to 15 Mbps	Up to 100 Mbps	Up to 300 Mbps
Upload Speed	Up to 1 Mbps	Up to 30 Mbps	Up to 100 Mbps
Technology	Older copper line technology	Improved DSL over short copper lines	Enhanced VDSL with vectoring
Line Quality Sensitivity	Very sensitive to line quality	Moderate	Improved noise cancellation

Feature	ADSL Router	VDSL Router	SuperVector Router
Distance Limitation	Performs poorly beyond 2 km	Perform well up to 1 km	Best performance under 500 meters
Latency	Higher latency	Lower latency	Very low latency
Use Case	Basic browsing, email	HD streaming, online gaming	4K streaming, smart home, heavy usage
Price	Lowest	Moderate	Highest

DSL Technical Troubleshooting

- Physical issue
- Logical issue

Physical issue

- Cutting in the wire
- Wrong Card
- Physical Instability

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- **Cutting in the wire:**
 - Data Down
 - Voice Down
 - Data & voice Down

- **Wrong Card:**
 - Two lines in the same port.
- **Physical Instability:**
 - cutting or rusted cable
 - Problem with the router's hardware

Logical issue
Managed by NOC team

- IP Conflict
- DNS Error
- DHCP

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- **IP Conflict:**
 - Two Device in the same IP
 - **DNS Error:**
 - When the Domain Name System fails to resolve a website address to its corresponding IP address.
 - **DHCP (Dynamic Host Configuration Protocol):**
 - When a device fails to obtain an IP address from the DHCP server, this usually results in no internet access and can be caused by server issues, network misconfigurations, or timeouts.

ONT, ONU FTTH Routers

Feature	ONT (Optical Network Terminal)	ONU (Optical Network Unit)
Function	Converts optical signal to electrical for home use	Similar function, often used in shared buildings

Feature	ONT (Optical Network Terminal)	ONU (Optical Network Unit)
Location	Typically installed inside the customer's home	May be placed outside or in shared hallway
Ports	Includes Ethernet, VoIP, and sometimes Wi-Fi	May have fewer ports or connect to a router
Power Supply	Requires power from customer premises	Usually powered in a shared cabinet
Integrated Router	Often includes built-in router/Wi-Fi	Usually requires separate router
Usage by WE	Common in individual homes with FTTH lines	Used in buildings with multiple users
Speed Support	Supports full fiber speeds (up to 1 Gbps+)	Also supports high fiber speeds

FTTH Technical Troubleshooting.

- **Physical Layer Checks.**
 - **Inspect Fiber & Connectors**
 - Clean all SC/APC or LC connectors with isopropyl wipes or specialized cleaning tools.
 - Check for bends, kinks or breaks along the cable route.
 - **Verify Optical Power Levels**

- Use an Optical Power Meter to confirm Rx (–28 to –8 dBm) and Tx levels per PON spec.
 - If power is below threshold ⇒ look for dirty connectors, excessive attenuation, or splitter fault.
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- **ONT/ONU Status**

- **LED Indicators**

- **PON** LED off or blinking ⇒ no sync with OLT (check fiber & power).
 - **LOS** (Loss of Signal) ON ⇒ fiber break or dirty connector.

- **Re-provisioning**

- Log into ONT's web GUI or ACS (TR-069) server, force a re-registration (Reboot/Re-bind).
 - Check SN/Password match the OLT's VLAN/PON port config.
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- **DHCP & IP Connectivity**

- **IP Address Assignment**

- Confirm ONT successfully obtains IP from DHCP server.
 - If no IP ⇒ check DHCP pools, option-82 settings, and VLAN tagging.

- **Ping & Traceroute**

- From ONT CLI or a PC behind it, ping the OLT's default gateway and public DNS (e.g., 8.8.8.8).
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MIFI, Wingle, WEair Mobile Internet Devices

Feature	MiFi	Wingle	WE Air
Device Type	Portable Wi-Fi hotspot	USB modem with hotspot feature	Fixed wireless home router
Portability	High – battery powered	Medium – USB powered (PC/laptop needed)	Low – intended for fixed home/office use
Power Source	Rechargeable battery	USB port	AC adapter
Internet Source	4G/3G SIM	4G/3G SIM	4G/4G+ SIM or outdoor antenna
Wi-Fi Sharing	supports up to 10–15 devices	supports up to 5–10 devices	supports 32+ devices
Speed Support	Up to 150 Mbps	Up to 150 Mbps	Up to 300 Mbps (depending on model)
Best For	On-the-go internet (travel, outdoors)	Laptop/PC users needing occasional internet	Home/office users needing stable high-speed
SIM Type	Micro/Nano SIM	Standard/Micro SIM	Micro SIM (may vary)
Additional Features	Portable, LCD screen, battery status	Plug & play, compact size	LAN ports, stronger signal, better coverage

Feature	MiFi	Wingle	WE Air
Price Range	Moderate	Budget-friendly	Higher – but more powerful

MiFi, Wingle, WEair Mobile Internet Troubleshooting

No Internet Connection

- **Check SIM Status**
Ensure the SIM card is inserted properly and is active (not expired or blocked).
- **Signal Strength**
Confirm sufficient 4G/3G signal bars. Try moving to an open area or near a window.
- **Data Balance**
Use the WE app or USSD code to check if there's remaining data quota.
- **Restart Device**
Power off the MiFi/Wingle/WE Air for 10 seconds, then turn it back on.

Device Not Detected (Wingle)

- **USB Port Issues**
Try a different USB port or another computer.
- **Driver Missing**
Install required drivers from the device's virtual CD-ROM or official WE website.
- **No Light Indicator**
If LED doesn't turn on, check power supply or device damage.

Slow Internet Speed

- **Network Congestion**

Test at a different time or location.

- **Connected Devices**

Too many users can slow performance—limit the number of devices connected.

- **Signal Quality**

Use the device web interface to check signal dBm level (ideal: –70 dBm to –90 dBm).

- **Frequency Band Locking**

In some advanced models, lock to 4G only or preferred band via settings (1800 MHz).

Wi-Fi Issues

- **SSID Not Visible**

Restart the device and check if Wi-Fi is turned on in the settings.

- **Wrong Password**

Reset password from the admin panel (usually 192.168.8.1 or printed on the back).

- **Interference**

Change the Wi-Fi channel to reduce interference from nearby networks.

WE Air – Specific Issues

- **Poor Indoor Coverage**

Use external antennas (if supported), or move the device near a window.

- **SIM Not Recognized**

Confirm SIM is data-enabled and supports LTE. Reinsert or try another SIM for testing.

ADSL, VDSL, Supervector Router Configuration.

ADSL Router – Huawei HG531 V1

1. Connect to the router:

- Connect your PC via LAN cable or Wi-Fi (SSID and password printed on label).
- Open browser → enter IP: 192.168.1.1.

2. Login:

- Username: admin
- Password: admin or found on router label.

3. Setup ADSL Internet:

- Go to **Basic** → **WAN**.
- Set:
 - **Username/Password:** (provided by WE)
 - **VPI/VCI:** 0 / 35 (default for Egypt)

4. Save & reboot.

5. Wi-Fi Settings (optional):

- Go to **WLAN** → **Basic**
- Change SSID, password (WPA2 recommended).

VDSL Router – ZTE ZXHN H168N

1. Connect via LAN or Wi-Fi.

- IP: 192.168.1.1

2. Login:

- Username: admin
- Password: On sticker (or admin)

3. Setup VDSL WAN:

- Go to **Network → WAN**
- Choose **WAN Type: VDSL**
- Enter WE username & password

4. Save → Reboot.

5. Change Wi-Fi Settings under Wireless → SSID Settings.

Supervector Router – Huawei DN8245V

Connect to router via LAN or Wi-Fi.

- Default IP: 192.168.1.1.

2. Login:

- Username: telecomadmin
- Password: On sticker

3. Set WAN Connection:

- Go to **Internet → WAN**
- Input WE username/password
- Apply settings

4. Wi-Fi Setup:

- Go to **WLAN → SSID Settings**
 - Set:
 - 2.4GHz and 5GHz names separately.
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MIFI, Wingle, WEair Configuration.

MiFi Router – Huawei E5577Cs-321

1. Connect to Wi-Fi:

- SSID and Wi-Fi password are printed on the back or shown on the screen.

2. Access Web Interface:

- Open your browser → go to 192.168.8.1
- Username: admin
- Password: admin (or printed on the device)

3. Internet Settings:

- Navigate to **Settings → Dial-up → Profile Management**
 - **Profile Name:** WE
 - **APN:** internet.te.eg
- Click Save, then Apply.

4. Wi-Fi Settings (Optional):

- Go to **Wi-Fi Settings** to change SSID and password.

5. Restart the MiFi after applying changes.

Wingle Router – Huawei E8372h-608

1. Plug the Wingle:

- Insert into a laptop, car charger, or power bank USB port.

2. **Connect via Wi-Fi:**

- SSID and password are written on the back label.

3. **Access Web Interface:**

- Open browser → go to 192.168.1.1
- Username: admin
- Password: admin

4. **Set APN:**

- Go to **Settings → Dial-up → Profile Management**
 - Profile Name: WE
 - APN: internet.te.eg
- Save → Apply.

5. **Wi-Fi Configuration:**

- Navigate to **WLAN Settings** to rename the SSID and change the password.

WE Air Router – ZTE MF253V

1. **Insert SIM & Power On:**

- Insert the 4G SIM card into the slot, then plug in and power on the router.

2. **Connect to the Router:**

- Use the default SSID and password (printed on the device) or LAN cable.

3. **Access Admin Interface:**

- Open browser → enter 192.168.1.1
- Username: admin
- Password: admin (or as printed on the label)

4. **Configure APN:**

- Go to **Settings → Network → APN Settings**
 - APN: internet.te.eg
 - Mode: Auto
- Save and Apply.

5. **Wi-Fi Configuration:**

- Navigate to **Wireless Settings** → change SSID and password as needed.

6. **Optimize Signal:**

- Place the router near a window or higher position for better 4G signal strength.

ONT, ONU FTTH Routers Configuration.

ONT Router – Huawei EchoLife HG8245H

1. Connect to the ONT:

- Connect your PC to the ONT via LAN cable or Wi-Fi.
- Open your browser and go to: 192.168.100.1

2. Login:

- **Username:** telecomadmin
- **Password:** Usually on the sticker (admintelecom or admin@Huawei123)

3. Internet Setup (PPPoE):

- Go to **WAN → WAN Configuration**

- Set:
 - **WAN Mode:** Route WAN
 - **Connection Type:** PPPoE
 - **VLAN:** Enable → ID: 835
 - Enter **username & password** provided by WE (user@te.eg)
 - 4. **Wi-Fi Settings:**
 - Go to **WLAN → 2.4GHz / 5GHz**
 - Change **SSID** (network name) and **Wi-Fi password**
 - Use **WPA2-PSK** or **WPA3** if available for best security
 - 5. **Save & Reboot**
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ONU Device – ZTE ZXHN F660 (used as ONU)

Typically used in apartment buildings or multi-tenant fiber connections, where it acts as a bridge between fiber and a home router.

1. **Connect to the ONU:**
 - Usually connected to a router via Ethernet.
 - If accessible directly, connect via LAN and access 192.168.1.1
2. **Login:**
 - **Username:** admin
 - **Password:** admin or from label
3. **Bridge Mode Configuration:**
 - Go to **Network → WAN → WAN Connection**
 - Set:
 - **Connection Type:** Bridge
 - **VLAN ID:** 835
 - Save settings
4. **Use External Router (TP-Link, Huawei):**
 - Connect ONU LAN port to WAN port of the external router
 - Configure **PPPoE** on the router with WE credential
 - Set up Wi-Fi, DHCP, and NAT on the external router
5. **Save & Reboot Both Devices**

important terms.

- CPE : Customer Premises Equipment.
 - BLQ : Bad Line Quality.
 - STB : Receiver which responsible for IPTV.
 - DHCP: Dynamic Host Configuration Protocol.
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